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Integration Reference Point (IRP) overview and usage guide
(3GPP TS 32.103 version 12.3.0 Release 12)**



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Foreword

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Introduction

This specification provides the IRP Overview for 3GPP specifications, addressing Interface IRPs, NRM IRPs, Measurement / Trace /SON specifications, as well as their relationships with each other.

1 Scope

The present document gives an overview about 3GPP's management interface capabilities and related functionality. It provides high-level information about IRP Framework, available IRPs as well as their relationship with each other. This document is intended to be a guide into the world of 3GPP management specifications, enabling also non-subject matter experts to gain an understanding about 3GPP's management solutions.

In addition Annex A provides IRP packaging suggestions, intended to guide service providers as well as equipment and solutions providers to identify and choose suitable standardized management interface capabilities.

2 References

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[1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

[2] 3GPP TS 32.150: "Telecommunication management; Integration Reference Point (IRP) Concept and definitions".

[3] 3GPP TS 32.101: "Telecommunication management; Principles and high level requirements".

[4] 3GPP TS 32.102: "Telecommunication management; Architecture".

[5] 3GPP TS 32.111-1: "Telecommunication management; Fault Management; Part 1: 3G fault management requirements".

[6] 3GPP TS 32.111-2: "Telecommunication management; Fault Management; Part 2: Alarm Integration Reference Point (IRP): Information Service (IS)".

[7] 3GPP TS 32.111-6: "Telecommunication management; Fault Management; Part 6: Alarm Integration Reference Point (IRP): Solution Set (SS) definitions".

[8] 3GPP TS 32.121: "Telecommunication management; Advanced Alarm Management (AAM) Integration Reference Point (IRP): Requirements".

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[11] 3GPP TS 32.140: "Telecommunication management; Subscription Management (SuM) requirements".

[12] 3GPP TS 32.141: "Telecommunication management; Subscription Management (SuM) architecture".

[13] 3GPP TS 32.151: "Telecommunication management; Integration Reference Point (IRP) Information Service (IS) template".

[14] 3GPP TS 32.152: "Telecommunication management; Integration Reference Point (IRP) Information Service (IS) Unified Modelling Language (UML) repertoire".

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[17] 3GPP TS 32.155: "Telecommunication management; Requirements template".

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[20] 3GPP TS 32.176: Void

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[22] 3GPP TS 32.301: "Telecommunication management; Configuration Management (CM); Notification Integration Reference Point (IRP); Requirements".

[23] 3GPP TS 32.302: "Telecommunication management; Configuration Management (CM); Notification Integration Reference Point (IRP); Information Service (IS)".

[24] 3GPP TS 32.306: "Telecommunication management; Configuration Management (CM); Notification Integration Reference Point (IRP): Solution Set (SS) definitions".

[25] 3GPP TS 32.311: "Telecommunication management; Generic Integration Reference Point (IRP) management; Requirements".

[26] 3GPP TS 32.312: "Telecommunication management; Generic Integration Reference Point (IRP) management; Information Service (IS)".

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[28] 3GPP TS 32.321: "Telecommunication management; Test management Integration Reference Point (IRP); Requirements".

[29] 3GPP TS 32.322: "Telecommunication management; Test management Integration Reference Point (IRP); Information Service (IS) ".

[30] 3GPP TS 32.326: "Telecommunication management; Test management Integration Reference Point (IRP): Solution Set (SS) definitions".

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[33] 3GPP TS 32.336: "Telecommunication management; Notification Log (NL) Integration Reference Point (IRP): Solution Set (SS) definitions".

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[37] 3GPP TS 32.351: "Telecommunication management; Communication Surveillance (CS) Integration Reference Point (IRP); Requirements".

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[40] 3GPP TS 32.361: "Telecommunication management; Entry Point (EP) Integration Reference Point (IRP); Requirements".

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[42] 3GPP TS 32.366: "Telecommunication management; Entry Point (EP) Integration Reference Point (IRP); Solution Set (SS) definitions".

[43] 3GPP TS 32.381: "Telecommunication management; Partial Suspension of Itf-N Integration Reference Point (IRP); Requirements".

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[46] 3GPP TS 32.391: "Telecommunication management; Delta synchronization Integration Reference Point (IRP); Requirements".

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[48] 3GPP TS 32.396: "Telecommunication management; Delta synchronization Integration Reference Point (IRP); Solution Set (SS) definitions".

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[53] 3GPP TS 32.408: "Telecommunication management; Performance Management (PM); Performance measurements; Teleservice".

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[76] 3GPP TS 32.501: "Telecommunication management; Self-configuration of network elements; Concepts and requirements".

[77] 3GPP TS 32.502: "Telecommunication management; Self-configuration of network elements Integration Reference Point (IRP); Information Service (IS)".

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[89] 3GPP TS 32.601: "Telecommunication management; Configuration Management (CM); Basic CM Integration Reference Point (IRP); Requirements".

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[92] 3GPP TS 32.611: "Telecommunication management; Configuration Management (CM); Bulk CM Integration Reference Point (IRP): Requirements".

[93] 3GPP TS 32.612: "Telecommunication management; Configuration Management (CM); Bulk CM Integration Reference Point (IRP): Information Service (IS)".

[94] 3GPP TS 32.616: "Telecommunication management; Configuration Management (CM); Bulk CM Integration Reference Point (IRP); Solution Set (SS) definitions".

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[96] 3GPP TS 32.622: "Telecommunication management; Generic Network Resource Model (NRM) Integration Reference Point (IRP); Information Service (IS)".

[97] 3GPP TS 32.626: "Telecommunication management; State management data definition Integration Reference Point (IRP); Solution Set (SS) definitions".

[98] 3GPP TS 32.631: "Telecommunication management; Configuration Management (CM); Core network resources Integration Reference Point (IRP); Requirements".

[99] 3GPP TS 32.632: "Telecommunication management; Configuration Management (CM); Core Network Resources Integration Reference Point (IRP); Network Resource Model (NRM)".

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[101] 3GPP TS 32.641: "Telecommunication management; Configuration Management (CM); UTRAN network resources Integration Reference Point (IRP); Requirements".

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[108] 3GPP TS 32.662: "Telecommunication management; Configuration Management (CM); Kernel CM Information Service (IS)".

[109] 3GPP TS 32.671: "Telecommunication management; Configuration Management (CM); State Management Integration Reference Point (IRP); Requirements".

[110] 3GPP TS 32.672: "Telecommunication management; Configuration Management (CM); State Management Integration Reference Point (IRP); Information Service (IS)".

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[113] 3GPP TS 32.722: "Telecommunication management; Configuration Management (CM); Repeater network resources Integration Reference Point (IRP); information Service (IS)".

[114] 3GPP TS 32.726: "Telecommunication management; Configuration Management (CM); Repeater network resources Integration Reference Point (IRP); Solution Set (SS) definitions".

[115] 3GPP TS 32.761: "Telecommunication management; Evolved Universal Terrestrial Radio Access Network (E-UTRAN) Network Resource Model (NRM) Integration Reference Point (IRP); Requirements".

[116] 3GPP TS 32.762: "Telecommunication management; Evolved Universal Terrestrial Radio Access Network (E-UTRAN) Network Resource Model (NRM) Integration Reference Point (IRP); Information Service (IS)".

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[118] 3GPP TS 52.402: "Telecommunication management; Performance Management (PM); Performance measurements – GSM".

[119] 3GPP TS 32.181: "Telecommunication management; User Data Convergence (UDC); Framework for Model Handling and Management".

[120] 3GPP TS 32.182: "Telecommunication management; User Data Convergence (UDC); Common baseline information model (CBIM)".

[121] 3GPP TS 28.620: "Telecommunication management; Fixed Mobile Convergence (FMC) Federated Network Information Model (FNIM) Umbrella Information Model (UIM)".

[122] 3GPP TS 28.621: "Telecommunication management; Generic Network Resource Model (NRM) Integration Reference Point (IRP); Requirements".

[123] 3GPP TS 28.622: "Telecommunication management; Generic Network Resource Model (NRM) Integration Reference Point (IRP); Information Service (IS)".

[124] 3GPP TS 28.623: "Telecommunication management; Generic Network Resource Model (NRM) Integration Reference Point (IRP); Solution Set (SS) definitions".

[125] 3GPP TS 28.627: "Telecommunication management; Self-Organizing Networks (SON) Policy Network Resource Model (NRM) Integration Reference Point (IRP); Requirements".

[126] 3GPP TS 28.628: "Telecommunication management; Self-Organizing Networks (SON) Policy Network Resource Model (NRM) Integration Reference Point (IRP); Information Service (IS) ".

[127] 3GPP TS 28.629: "Telecommunication management; Self-Organizing Networks (SON) Policy Network Resource Model (NRM) Integration Reference Point (IRP); Solution Set (SS) definitions".

[128] 3GPP TS 28.631: "Telecommunication management; Inventory Management (IM) Network Resource Model (NRM) Integration Reference Point (IRP); Requirements".

[129] 3GPP TS 28.632: "Telecommunication management; Inventory Management (IM) Network Resource Model (NRM) Integration Reference Point (IRP); Information Service (IS)".

[130] 3GPP TS 28.633: "Telecommunication management; Inventory Management (IM) Network Resource Model (NRM) Integration Reference Point (IRP); Solution Set (SS) definitions".

[131] 3GPP TS 28.651: "Telecommunication management; Universal Terrestrial Radio Access Network (UTRAN) Network Resource Model (NRM) Integration Reference Point (IRP); Requirements".

[132] 3GPP TS 28.652: "Telecommunication management; Universal Terrestrial Radio Access Network (UTRAN) Network Resource Model (NRM) Integration Reference Point (IRP); Information Service (IS)".

[133] 3GPP TS 28.653: "Telecommunication management; Universal Terrestrial Radio Access Network (UTRAN) Network Resource Model (NRM) Integration Reference Point (IRP); Solution Set (SS) definitions".

[134] 3GPP TS 28.654: "Telecommunication management; GSM/EDGE Radio Access Network (GERAN) Network Resource Model (NRM) Integration Reference Point (IRP); Requirements".

[135] 3GPP TS 28.655: "Telecommunication management; GSM/EDGE Radio Access Network (GERAN) Network Resource Model (NRM) Integration Reference Point (IRP); Information Service (IS)".

[136] 3GPP TS 28.656: "Telecommunication management; GSM/EDGE Radio Access Network (GERAN) Network Resource Model (NRM) Integration Reference Point (IRP); Solution Set (SS) definitions".

[137] 3GPP TS 28.657: "Telecommunication management; Evolved Universal Terrestrial Radio Access Network (E-UTRAN) Network Resource Model (NRM) Integration Reference Point (IRP); Requirements".

[138] 3GPP TS 28.658: "Telecommunication management; Evolved Universal Terrestrial Radio Access Network (E-UTRAN) Network Resource Model (NRM) Integration Reference Point (IRP); Information Service (IS)".

[139] 3GPP TS 28.659: "Telecommunication management; Evolved Universal Terrestrial Radio Access Network (E-UTRAN) Network Resource Model (NRM) Integration Reference Point (IRP); Solution Set (SS) definitions".

[140] 3GPP TS 28.661: "Telecommunication management; Generic Radio Access Network (RAN) Network Resource Model (NRM) Integration Reference Point (IRP); Requirements".

[141] 3GPP TS 28.662: "Telecommunication management; Generic Radio Access Network (RAN) Network Resource Model (NRM) Integration Reference Point (IRP); Information Service (IS)".

[142] 3GPP TS 28.663: "Telecommunication management; Generic Radio Access Network (RAN) Network Resource Model (NRM) Integration Reference Point (IRP); Solution Set (SS) definitions".

[143] 3GPP TS 28.671: "Telecommunication management; Home Node B (HNB) Subsystem (HNS) Network Resource Model (NRM) Integration Reference Point (IRP); Requirements".

[144] 3GPP TS 28.672: "Telecommunication management; Home Node B (HNB) Subsystem (HNS) Network Resource Model (NRM) Integration Reference Point (IRP); Information Service (IS)".

[145] 3GPP TS 28.673: "Telecommunication management; Home Node B (HNB) Subsystem (HNS) Network Resource Model (NRM) Integration Reference Point (IRP); Solution Set (SS) definitions".

[146] 3GPP TS 28.674: "Telecommunication management; Home enhanced Node B (HeNB) Subsystem (HeNS) Network Resource Model (NRM) Integration Reference Point (IRP); Requirements".

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[149] 3GPP TS 28.701: "Telecommunication management; Core Network (CN) Network Resource Model (NRM) Integration Reference Point (IRP); Requirements".

[150] 3GPP TS 28.702: "Telecommunication management; Core Network (CN) Network Resource Model (NRM) Integration Reference Point (IRP); Information Service (IS)".

[151] 3GPP TS 28.703: "Telecommunication management; Core Network (CN) Network Resource Model (NRM) Integration Reference Point (IRP); Solution Set (SS) definitions".

[152] 3GPP TS 28.704: "Telecommunication management; IP Multimedia Subsystem (IMS) Network Resource Model (NRM) Integration Reference Point (IRP); Requirements".

[153] 3GPP TS 28.705: "Telecommunication management; IP Multimedia Subsystem (IMS) Network Resource Model (NRM) Integration Reference Point (IRP); Information Service (IS)".

[154] 3GPP TS 28.706: "Telecommunication management; IP Multimedia Subsystem (IMS) Network Resource Model (NRM) Integration Reference Point (IRP); Solution Set (SS) definitions".

[155] 3GPP TS 28.707: "Telecommunication management; Evolved Packet Core (EPC) Network Resource Model (NRM) Integration Reference Point (IRP); Requirements".

[156] 3GPP TS 28.708: "Telecommunication management; Evolved Packet Core (EPC) Network Resource Model (NRM) Integration Reference Point (IRP); Information Service (IS)".

[157] 3GPP TS 28.709: "Telecommunication management; Evolved Packet Core (EPC) Network Resource Model (NRM) Integration Reference Point (IRP); Solution Set (SS) definitions".

[158] 3GPP TS 28.731: "Telecommunication management; Transport Network (TN) interface Network Resource Model (NRM) Integration Reference Point (IRP); Requirements".

[159] 3GPP TS 28.732: "Telecommunication management; Transport Network (TN) interface Network Resource Model (NRM) Integration Reference Point (IRP); Information Service (IS)".

[160] 3GPP TS 28.733: "Telecommunication management; Transport Network (TN) interface Network Resource Model (NRM) Integration Reference Point (IRP); Solution Set (SS) definitions".

[161] 3GPP TS 28.734: "Telecommunication management; Signalling Transport Network (STN) interface Network Resource Model (NRM) Integration Reference Point (IRP); Requirements".

[162] 3GPP TS 28.735: "Telecommunication management; Signalling Transport Network (STN) interface Network Resource Model (NRM) Integration Reference Point (IRP); Information Service (IS)".

[163] 3GPP TS 28.736: "Telecommunication management; Signalling Transport Network (STN) interface Network Resource Model (NRM) Integration Reference Point (IRP); Solution Set (SS) definitions".

[164] 3GPP TS 28.751: "Telecommunication management; Subscription Management (SuM) Network Resource Model (NRM) Integration Reference Point (IRP); Requirements".

[165] 3GPP TS 28.752: "Telecommunication management; Subscription Management (SuM) Network Resource Model (NRM) Integration Reference Point (IRP); Information Service (IS)".

[166] 3GPP TS 28.753: "Telecommunication management; Subscription Management (SuM) Network Resource Model (NRM) Integration Reference Point (IRP); Information Service (IS)".

[167] 3GPP TS 32.107: "Telecommunication management; Fixed Mobile Convergence (FMC) Federated Network Information Model (FNIM)".

3 Definitions and abbreviations

3.1 Definitions

For the purposes of the present document, the terms and definitions given in TR 21.905 [1], TS 32.150 [2].

3.2 Abbreviations

For the purposes of the present document, the abbreviations given in TR 21.905 [1] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in TR 21.905 [1].

FMC	Fixed Mobile Convergence
FNIM	Federated Network Information Model
IF	Interface
IRP	Integration Reference Point
IS	Information Service
SS	Solution Set
UIM	Umbrella Information Model

4 IRP Framework

4.1 Introduction

For the purpose of management interface development 3GPP has developed an interface concept known as Integration Reference Point (IRP) to promote the wider adoption of standardized management interfaces in telecommunication networks. The IRP concept and associated methodology employs model-driven development, protocol and technology neutral modelling methods as well as protocol specific solution sets to achieve its goals.

IRP Framework and Methodology related 3GPP specifications:

- 32.150 [2]: Integration Reference Point (IRP) Concept and definitions
- 32.151 [13]: IRP Information Service (IS) template (*applicable to 32-series IRPs*)
- 32.152 [14]: IRP IS Unified Modeling Language (UML) repertoire
- 32.153 [15]: IRP technology specific templates, rules and guidelines
- 32.154 [16]: Backward and Forward Compatibility Concept and definitions
- 32.155 [17]: Requirements template
- 32.156 [18]: Fixed Mobile Convergence (FMC) Model repertoire
- 32.157 [19]: IRP Information Service (IS) template (*applicable for FMC*)

4.2 IRP Framework Highlights

4.2.1 IRP Concept

IRP specifications are specified using a 3-level approach: Requirements, IS-level and SS-level.

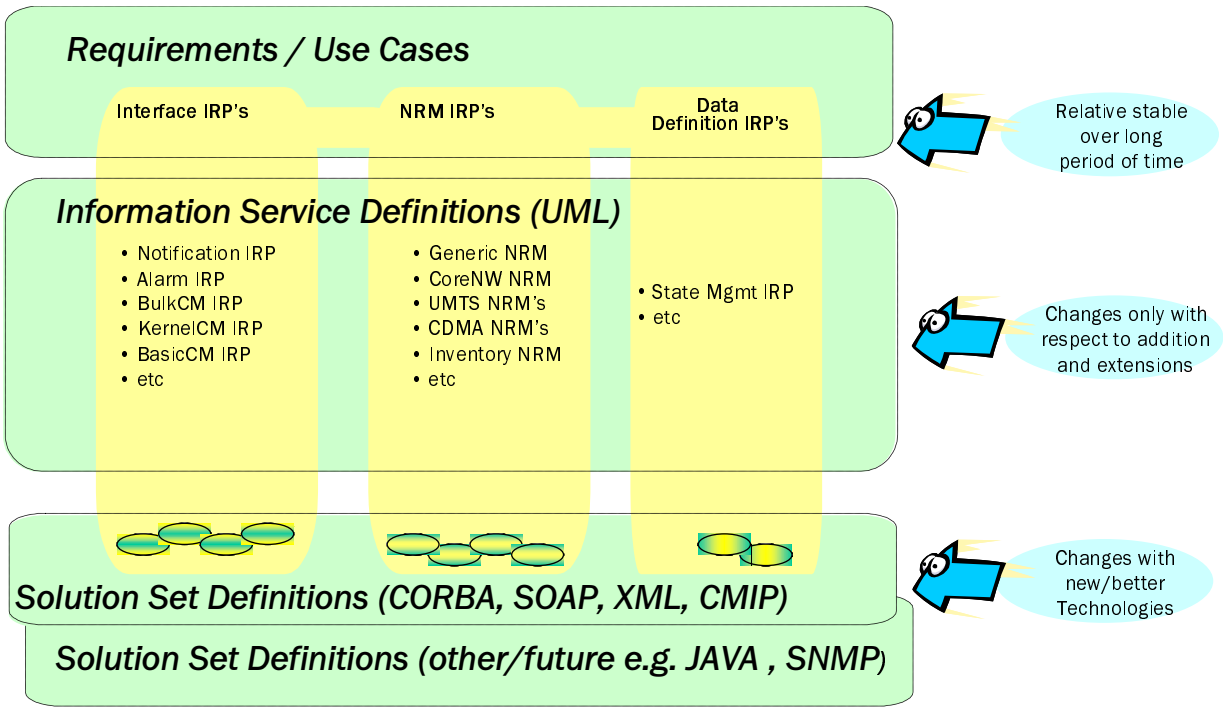


Figure 4.2-1: The IRP 3-Level Specifications Approach combined with the three IRP categories [2].

Requirements-level:

The "Requirements-level" intends to provide conceptual and use cases definitions for a specific management interface aspect as well as defining subsequent requirements for this IRP.

IS-level:

The "IS-level" provides the technology independent specification of an IRP.

SS-level:

The "SS-level" finally provides the mapping of IS definitions into one or more technology-specific Solution Sets. This concept provides support for multiple interface technologies as applicable on a vendor and/or network type basis and also enables accommodation of future interface technologies - without the need to redefine requirements and IS-level definitions.

4.2.2 Relationships between IRPs

There are three categories of IRP specifications:

- Interface IRPs
- NRM IRPs
- Data Definition IRPs

Each category is partitioned into Requirements, IS-level and SS-level specifications.

Relationships between various kinds of IRPs:

- Some Interface IRPs and NRM IRPs are used together. Such Interface IRPs operate on entities modelled by NRM IRPs. For example, operations defined in Basic CM IRP are used together with E- UTRAN NRM IRP to support E-UTRAN configuration management function.
- Interface IRPs are network/radio technology independent and can operate on entities modelled by NRM IRPs of different network technologies. For example, operations defined in Basic CM IRP are used together with E-UTRAN NRM IRP or UTRAN NRM IRP to support E- UTRAN or UTRAN configuration management functions.
- Interface IRPs are Converged Management ready (e.g. support management of fixed and/or mobile) and can operate on modelled mobile and/or fixed network entities.
- A Data Definition IRP provides common data definitions, referenced by specifications of Interface IRPs and NRM IRPs.

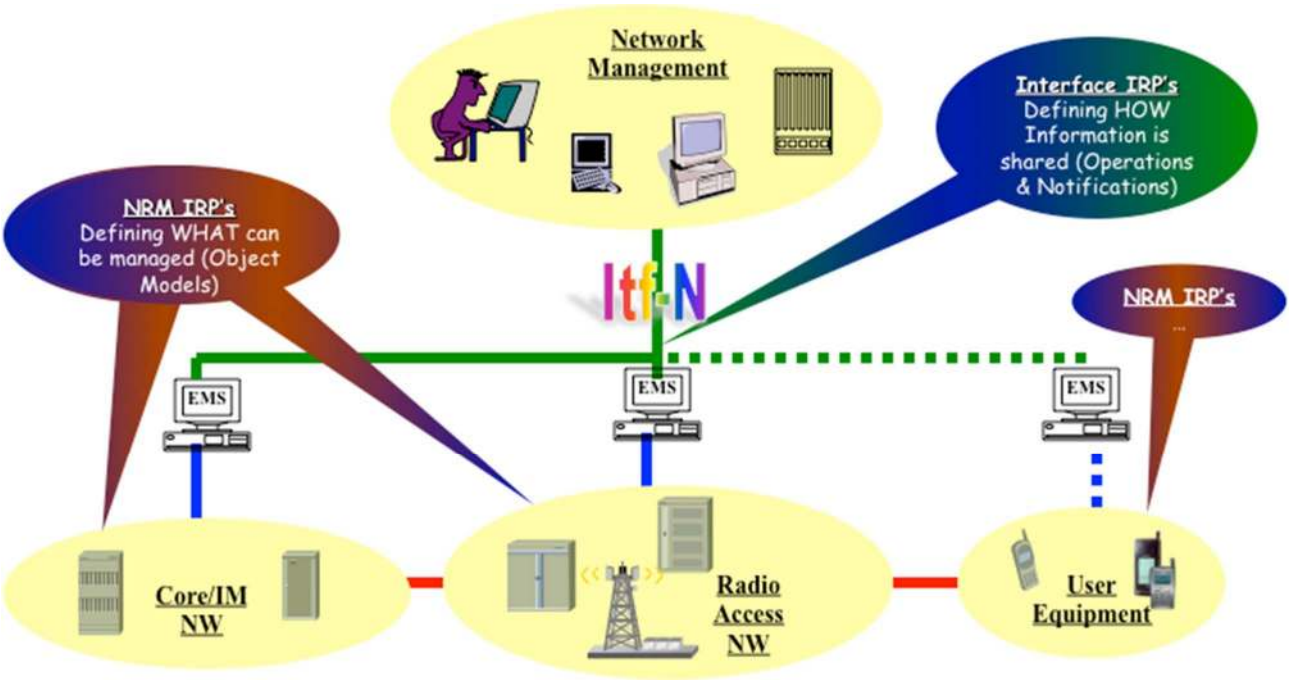


Figure 4.2-2: Relationship Interface IRP vs NRM IRP

4.2.3 IRP Development Principles

NRM IRP Extendibility - Enabling Technology, Organizational & Vendor-specific NRM extensions via

- Rule-based NRM Extensions (e.g. Sub-classing)
- vsDataContainer

Interface IRP Flexibility - Enabling: wide applicability, phased introduction capabilities & broad industry adoption (not just for wireless) via

- Flexible use of qualifiers "mandatory", "optional", "conditional" for operation, notifications and/or parameters
- NRM/Technology-neutrality & avoiding competing procedures

4.2.4 IRP Specification Structure

32-series IRPs follow a uniform specification structure:

- xx1: xyz IRP Requirements
- xx2: xyz IRP IS
- *xx3: xyz IRP CORBA SS (merged into xx6 after R9)*
- *xx4: xyz IRP CMIP SS (retired after R6)*
- *xx5: xyz IRP XML Definitions (applicable for NRM's & notification emitting IRPs - merged into xx6 after R9)*
- xx6: xyz IRP SS (since R10, containing all technology specific Solution Set definitions)

  xx7: xyz IRP SOAP SS (merged into xx6 after R9)

28-series IRPs (supporting FMC) consisting of 3-specification sets, structured as follows:

- 28.3xx: Interface IRP specifications.
- 28.620-28.649: Common/Generic NRM IRPs.
- 28.650-28.699: Access Network related NRM IRPs.
- 28.700-28.729: CN-related NRM IRPs.
- 28.730-28.749: Transport-related NRM IRPs.
- 28.750-28.769: Service-related NRM IRPs.
- *28.770-28.799: reserved for future topic NRM IRPs.*

4.2.5 Essential IRP Definitions [2]

Data Definition IRP: 3GPP publishes IRP specifications relating to commonly used data definitions that can be imported for use by Interface IRP and/or NRM IRP. This term represents all such specifications. An example of a Data Definition IRP is the State Management IRP (32.671 [109], 32.672 [110], 32.676 [111], etc).

Integration Reference Point (IRP): An architectural concept that is described by a set of specifications for definition of a certain aspect of a management interface, comprising a **Requirements** specification, an **Information Service** specification, and one or more **Solution Set** specifications.

Interface IRP: 3GPP publishes a number of IRP specifications each of which is related to a set of operations and notifications for a specific telecom management domain such as alarm management, configuration management, etc. Interface IRPs also contain definitions of Support IOCs. This term represents all such specifications. An example of an Interface IRP is the Basic CM IRP (the set of TSs 32.601 [89], 32.602 [90], 32.606 [91], etc.).

IRPAgent: Encapsulates a well-defined subset of network (element) functions. It interacts with IRPManagers using one or more IRPs. From the IRPManager's perspective, the IRPAgent behaviour is only visible via the IRP(s).

Information Service (IS): an IRP Information Service describes the information related to the entities (either network resources or support objects) to be managed and the way that the information may be managed for a certain functional area (e.g. the Alarm IRP Information Service in the fault management area). Information Services are defined for all IRPs.

IRPManager: Models a user of IRPAgent(s) and it interacts directly with the IRPAgent(s) using IRP(s).
Since the IRPManager represents an IRPAgent user, it gives a clear picture of what the IRPAgent is supposed to do. From the IRPAgent perspective, the IRPManager behaviour is only visible via the IRP.

Network Resource Model (NRM): An Information Service describing Information Object Classes representing the manageable aspects of network resources, e.g. an RNC or NodeB.

NRM IRP: 3GPP publishes a number of IRP specifications each of which is related to a particular NRM (Network Resource Model) as defined in 3GPP TS 32.101 [3]. NRM IRPs do not define any operations or notifications. This term represents all such specifications. Note: In some NRM IRP titles, for historic reasons, they are named "…network resources IRP"). An example of an NRM IRP is the Generic NRM IRP (32.621 [95], 32.622 [96], 32.626 [97], etc.).

Solution Set (SS): contains a mapping of the IRP Information Service (IS) defined entities (that are technology-agnostic) to technology specific termed entities. It does not contain specification of the entities' semantics. The semantics can be found in the corresponding IS. It is noted that one IS can be mapped to one or several SSs.

4.3 IRP related High-level Requirements & Architecture Specifications

The IRP Framework has been developed in response to management requirements documented in the following specifications:

- 32.101 [3]: Principles and high level requirements

- 32.102 [4]: Architecture
- 32.107 [167]: Fixed Mobile Convergence (FMC) Federated Network Information Model (FNIM)
- 32.111-1 [5]: 3G fault management requirements
- 32.140 [11],32.141 [12]: SuM requirements / SuM architecture
- 32.300 [21]: Name convention for Managed Objects
- 32.401 [49]: PM Concept and requirements
- 32.500 [75]: SON Concepts and requirements
- 32.511 [79]: ANR management Concepts and requirements
- 32.521 [80]: Self-optimization Concepts and requirements
- 32.541 [86]: Self-healing Concepts and requirements
- 32.551 [87]: Energy Saving Management (ESM) Concepts and requirements
- 32.600 [88]: CM Concept and high-level requirements

4.4 Converged Management and IRPs

As outlined in previous sections, 3GPP Interface IRPs are network-technology-independent, and therefore can be utilized for the management of various network technologies.

To further ease converged management (the common management of multiple network technologies) and to address the need for OPEX reduction, 3GPP together with other industry organizations (i.e. NGMN and TM Forum) developed jointly the following specifications:

- Fixed Mobile Convergence (FMC) Federated Network Information Model (FNIM) (published in 3GPP as TS 32.107 [167])
    - Defining conceptually the federated network information model and the relationships between participating models (see also section 5.5 of TS 32.101 [3]).
- FMC FNIM Umbrella Information Model (UIM) (published in 3GPP as TS 28.620[121])
    - Providing abstract definitions of information objects applicable across Domain/Technology-specific Concrete Models.
- Fixed Mobile Convergence (FMC) Model repertoire (published in 3GPP as TS 32.156[18])
    - Defining meta definitions for models supporting converged management.
- Fixed Mobile Convergence (FMC) 3GPP / TM Forum Concrete Model Relationships and Use Cases .
    - Proposing the specific structure of model and realisation of that structure across 3GPP and TM Forum by taking advantage of TM Forum transport-oriented model work (SID, MTNM/MTOSI) as well as 3GPP mobile-oriented model work (NRM IRPs etc.).

Based on above, 3GPP has applied definitions provided by these specifications by creating the 28-series variants of the 3GPP NRM IRPs to support converged management.

5 Interface IRPs

5.1 Interface IRP Overview

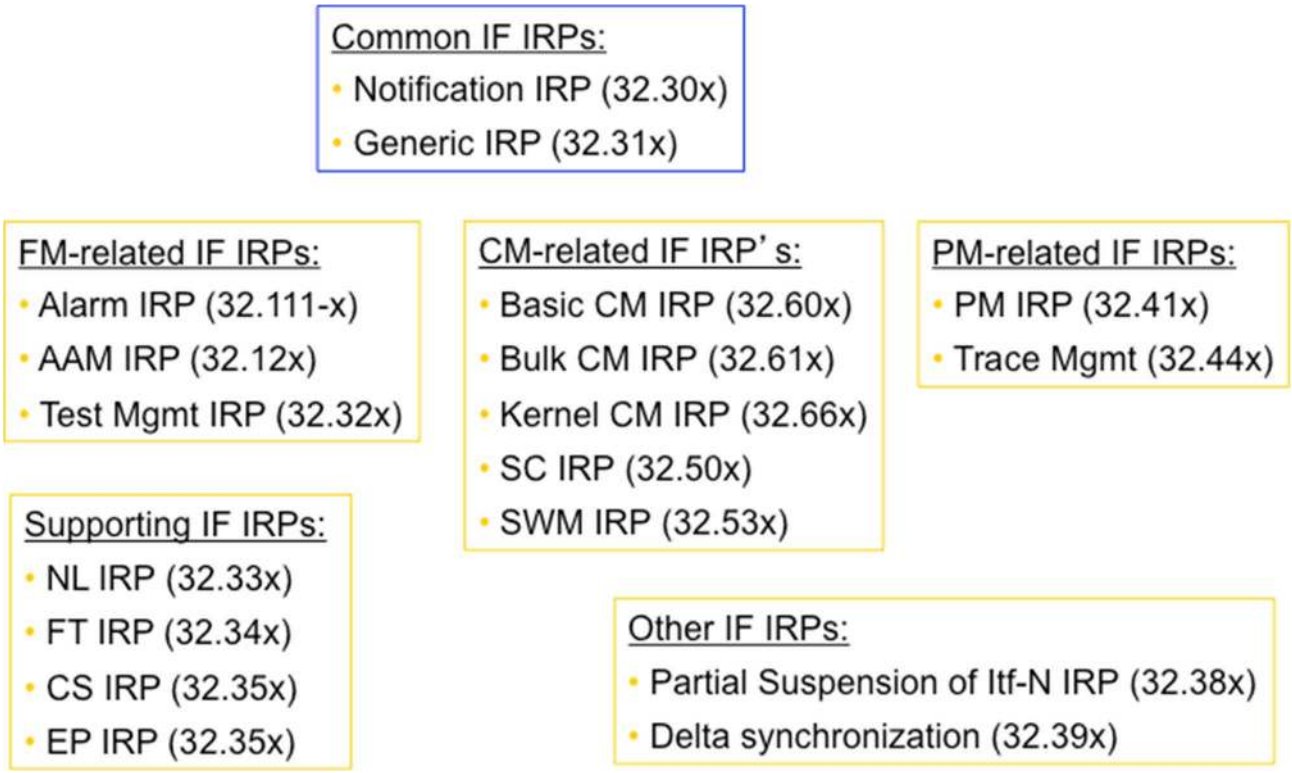


Figure 5.1-1: 3GPP Interface IRPs

5.2 Generic/Common Interface IRPs

Notification IRP – 32.300 [21], 32.301 [22], 32.302 [23], 32.306 [24]

The purpose of the Notification IRP is to define an interface through which an IRPManager can subscribe to an IRPAgent for receiving notifications, manage these subscriptions, and optionally set filters.

Generic IRP – 32.311 [25], 32.312 [26], 32.316 [27]

This IRP defines common services supported by all IRPs such as AlarmIRP. With this common service supported by all IRPs, an IRPManager can retrieve the profile of operations and notifications supported by a given IRP name-contained by an IRPAgent. An IRPManager can also retrieve the IRPVersions supported by a given IRP.

5.3 FM related Interface IRPs

Alarm IRP – 32.111-1 [5], 32.111-2 [6], 32.111-6 [7]

The Alarm (IRP) addresses the alarm surveillance aspects of Fault Management (FM), applied to Itf-N. It defines alarm semantics & syntax as well as alarm management related operations and notifications.

Advanced Alarm Management IRP – 32.121 [8], 32.122 [9], 32.126 [10]

The purpose of Advanced Alarm Management (AAM) IRP is to define an interface through which an IRPManager can categorize alarm notifications. It defines, for the purpose of categorizing alarm notifications, the information observable and controlled by management system's client and it also specifies the semantics of the interactions used to carry this information.

Test management IRP – 32.321 [28], 32.322 [29], 32.326 [30]

The purpose of the Test management IRP is to define an interface through which an IRPManager can manage and monitor tests in NE's, as well as methods to receive test results.

5.4 CM related Interface IRPs

Self-Configuration IRP – 32.500 [75], 32.501 [76], 32.502 [77], 32.506 [78]

The Self-Configuration provides methods to allow automatic configuration of eNBs, which are brought newly into a network.

Software Management IRP – 32.531 [83], 32.532 [84], 32.536 [85]

The Software Management IRP provides an IRPManager with capabilities to automate software management via Itf-N.

Basic CM IRP – 32.600 [88], 32.601 [89], 32.602 [90], 32.606 [91]

Using this IRP, an IRPAgent can communicate basic Configuration Management related information to one or several IRPManagers. The function of this Basic CM IRP Information Service is to define an interface for the retrieval and modification of Configuration Management Information.

Bulk CM IRP – 32.611 [92], 32.612 [93], 32.616 [94]

Using the Bulk CM IRP an IRPAgent can communicate CM related information to one or several IRPManagers using bulk data transfer mehods.

- BulkCM SimpleUpload, enabling upload of resource information by the IRPManager without explicit session control
- BulkCM Controlled Upload, enabling a session controlled upload of resource information by the IRPManager
- BulkCM Controlled Upload & Provisioning, enabling a session controlled upload and provisioning of resource information by the IRPManager

Kernel CM IRP – 32.661 [107], 32.662 [108]

The function of this Kernel CM IRP Information Service is to define an interface that provides essential CM notification services. While it is not expected that the Kernel CM IRP alone will provide adequate CM capabilities, the Kernel CM IRP is expected to provide the common supporting CM notifications required for other CM IRPs such as the Basic CM IRP or the Bulk CM IRP, each of which require the Kernel CM IRP.

5.5 PM & Trace related Interface IRPs

Performance Management IRP – 32.410 [55] , 32.411 [56] , 32.412 [57] , 32.416 [58]

The Performance Management IRP is defines an interface through which an IRPManager can manage measurement jobs as well threshold values.

Trace Management IRP – 32.441 [67], 32.442 [68], 32.446 [69]

The Trace Management IRP is defines an interface through which an IRPManager can activate and manage equipment and subscriber trace jobs.

5.6 Special-purpose related Interface IRPs

Notification Log IRP – 32.331 [31], 32.332 [32], 32.326 [33]

The purpose of the Notification Log IRP is to define an interface through which an IRPManager can manage a Notification Log, as well as methods for retrieval of logged notifications.

File Transfer IRP – 32.341 [34], 32.342 [35], 32.346 [36]

The purpose of the File Transfer IRP is to define a generic tools set supporting the management of file transfers between IRPManager and IRPAgent. Curently the following management data types are supported by this IRP:

- performance data files
- configuration & inventory files
- test result files
- call trace files
- notification log files
- charging files

Communication Surveillance IRP – 32.351 [37], 32.352 [38], 32.356 [39]

The purpose of the Communication Surveillance IRP is to provide an IRPManager with the capability to monitor the health of the communications link towards an IRPAgent.

Entry Point IRP – 32.361 [40], 32.362 [41], 32.366 [42]

The purpose of the Entry Point IRP is to provide an IRPManager with discovery and entry point capabilities towards an IRPAgent and its supported IRPs.

Partial Suspension of Itf-N IRP – 32.381 [43], 32.382 [44], 32. 386[45]

The purpose of Partial Suspension of Itf-N IRP is to define an interface through which an IRPManager can suspend the forwarding of notifications via Itf-N which were generated in parts of the managed systems.

Delta synchronization IRP – 32.391 [46], 32.392 [47], 32.396 [48]

The purpose of Delta Synchronization IRP is to define an interface through which an IRPManager can request only those data that have changed (i.e. changed, were created or deleted) from a synchronization point onwards.

6 NRM IRPs

6.1 NRM IRP Overview

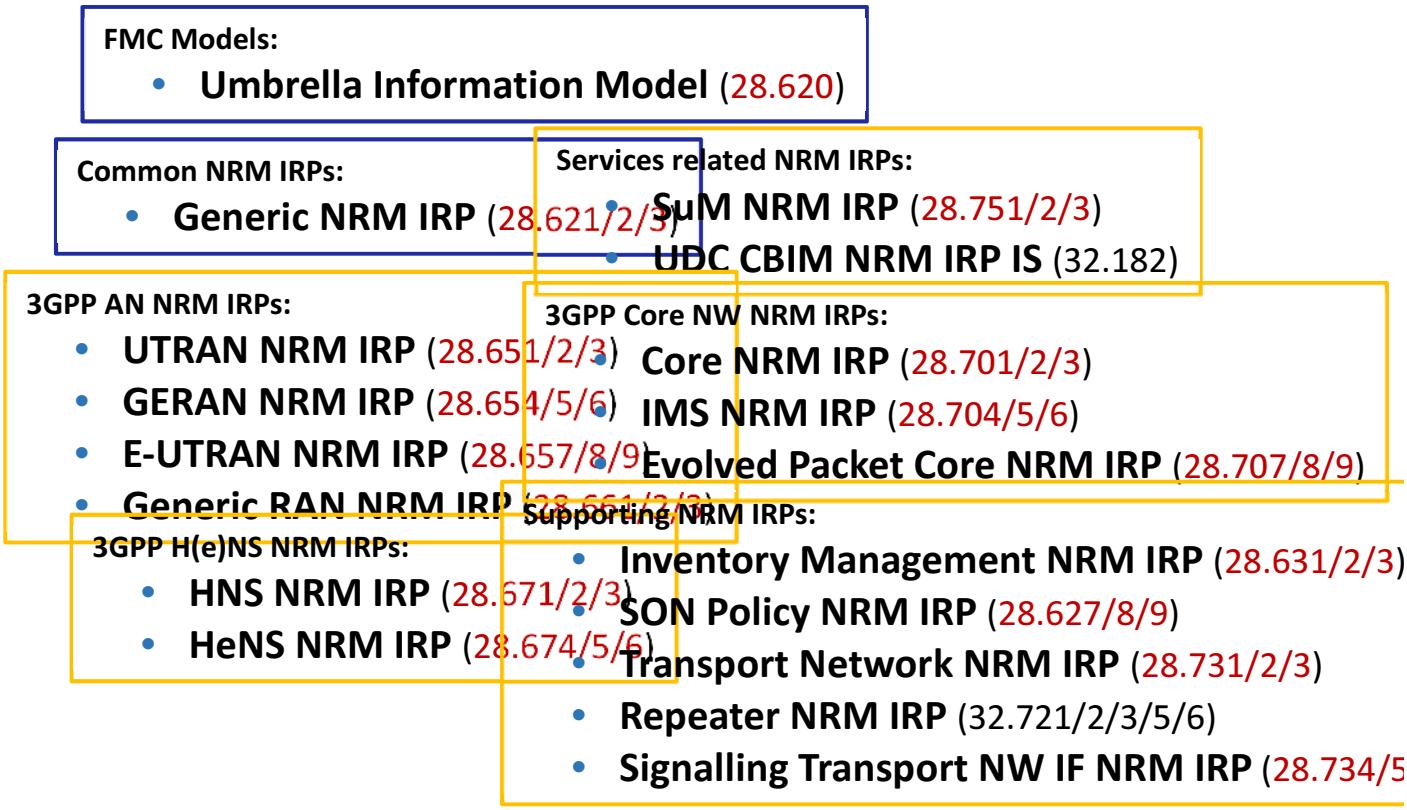


Figure 6.1-1: 3GPP NRM IRPs

6.2 FMC Models and Common NRM IRPs

FMC FNIM Umbrella Information Model (UIM) – 28.620 [121

The Umbrella Information Model (UIM) provides abstract definitions of information objects applicable across Domain/Technology-specific Concrete Models to enable end-to-end consistency of such definitions. These definitions are used by the Generic NRM IRP (28.621[122], 28.622 [123], 28.623 [124]) to enable 3GPP NRM IRPs to be part of the industry-wide Federated Network Information Model (FNIM).

Generic NRM IRP –28.621[122], 28.622 [123], 28.623 [124]

The Generic NRM IRP provides the generic network resources information that can be communicated between an IRPAgent and one or several IRPManagers for network management purposes. To enable converged management, the Generic NRM IRP as defined in 28.621[122], 28.622 [123], 28.623 [124] uses the common definitions specified in the FMC FNIM Umbrella Information Model (UIM).

6.3 Access Network (AN) related NRM IRPs

UTRAN NRM IRP –28.651 [131], 28.652 [132], 28.653 [133]

The UTRAN NRM IRP defines an IRP through which an IRPAgent can communicate Configuration Management information to one or several IRPManagers concerning UTRAN specific network resource, by reusing relevant parts of the Generic NRM IRP.

GERAN NRM IRP –28.654 [134], 28.655 [135], 28.656 [136]

The GERAN NRM IRP defines an IRP through which an IRPAgent can communicate Configuration Management information to one or several IRPManagers concerning GERAN specific network resource, by reusing relevant parts of the Generic NRM IRP.

E-UTRAN NRM IRP –28.657 [**137], 28.658 [138], 28.659 [139]**

The E-UTRAN NRM IRP defines an IRP through which an IRPAgent can communicate Configuration Management information to one or several IRPManagers concerning E-UTRAN specific network resource, by reusing relevant parts of the Generic NRM IRP.

Generic RAN NRM IRP –28.661 [140], 28.662 [141], 28.663 [142]

The Generic RAN NRM IRP defines an IRP through which an IRPAgent can communicate Configuration Management information to one or several IRPManagers concerning common RAN network resource (applicable to GERAN, UTRAN and E_UTRAN), by reusing relevant parts of the Generic NRM IRP.

6.4 Core/IMS related NRM IRPs

Core NRM IRP –28.701 [149], 28.702 [150], 28.703 [151]

The Core NRM IRP defines an IRP through which an IRPAgent can communicate Configuration Management information to one or several IRPManagers concerning core network specific resource, by reusing relevant parts of the Generic NRM IRP in.

IMS NRM IRP –28.704 [152], 28.705 [153], 28.706 [154]

The IMS NRM IRP defines an IRP through which an IRPAgent can communicate Configuration Management information to one or several IRPManagers concerning IP MultiMedia Subsystem specific network resource, by reusing relevant parts of the Generic NRM IRP.

Evolved Packet Core (EPC) NRM IRP –28.707 [155], 28.708 [156], 28.709 [157]

The EPC NRM IRP defines an IRP through which an IRPAgent can communicate Configuration Management information to one or several IRPManagers concerning Evolved Packet Core specific network resource, by reusing relevant parts of the Generic NRM IRP.

6.5 H(e)NB related NRM IRPs

HNS NRM IRP –28.671 [143], 28.672 [144], 28.673 [145]

The HNS NRM IRP defines an IRP through which an IRPAgent can communicate Configuration Management information to one or several IRPManagers concerning Home NodeB Subsystem specific network resource, by reusing relevant parts of the Generic NRM IRP.

HeNS NRM IRP –28.674 [146], 28.675 [147], 28.676 [148]

The HeNS NRM IRP defines an IRP through which an IRPAgent can communicate Configuration Management information to one or several IRPManagers concerning Home enhanced NodeB Subsystem specific network resource, by reusing relevant parts of the Generic NRM IRP.

6.6 Service Management related NRM IRPs and UDC specifications

SuM NRM IRP –28.751 [164], 28.752 [165], 28.753 [166]

Subscription Management is a feature that permits Service Providers, Value Added Service Providers and Mobile Operators to provision services for a specific subscriber. The SuM NRM IRP defines the Subscription Management specific network resource information between an IRPAgent and one or several IRPManagers.

UDC – 32.181 [119],**32.182** [120]

User Data Convergence (UDC) presents a layered architecture where user data is accessible at a logically unique repository called User Data Repository (UDR). The UDC CBIM defines a Common Baseline Information Model for UDC, which denotes an abstract, formal representation of entity types that are common to many applications using the UDR.

6.7 Special-purpose related NRM IRPs

Inventory Management NRM IRP –28.631 [128], 28.632 [129], 28.633 [130]

Inventory Management (IM) provides the operator with the ability to assure correct and effective operation of the telecom network as it evolves. IM actions have the objective to monitor the actual configuration on the Network Elements (NEs) and Network Resources (NRs). The Inventory Management NRM IRP defines the inventory specific network resource information between an IRPAgent and one or several IRPManagers.

SON Policy NRM IRP –28.627 [125], 28.628 [126], 28.629 [127]

The SON Policy NRM IRP defines an IRP through which an IRPAgent can communicate Configuration Management information to one or several IRPManagers concerning Self-Organizaing Networks Policies. Currently the following SON use cases are supported by this NRM IRP:

- SON Self-Optimization Management (requirements determined by TS 32.521[80], 28.627 [125])
- SON Self-Healing Management (requirements determined by TS 32.541[86])
- Energy Saving Management (requirements determined by TS 32.551 [87])

Transport Network NRM IRP –28.731 [158], 28.732 [159], 28.733 [160]

The Transport Network NRM IRP defines an IRP through which an IRPAgent can communicate Configuration Management information to one or several IRPManagers concerning Transport resources, by reusing relevant parts of the Generic NRM IRP.

Repeater NRM IRP – 32.721 [112], 32.722 [113], 32.726 [114]

The Repeater NRM IRP defines an IRP through which an IRPAgent can communicate Configuration Management information to one or several IRPManagers concerning Repeater resources, by reusing relevant parts of the Generic NRM IRP.

Signalling Transport NW IF NRM IRP –28.734 [161], 28.735 [162], 28.736 [163]

The Signalling Transport NW IF NRM IRP defines an IRP through which an IRPAgent can communicate Configuration Management information to one or several IRPManagers concerning Signalling Transport resources, by reusing relevant parts of the Generic NRM IRP.

7 Measurement & Trace Definitions

7.1 Performance Measurements

7.1.1 Overview and relationships to IRP Framework

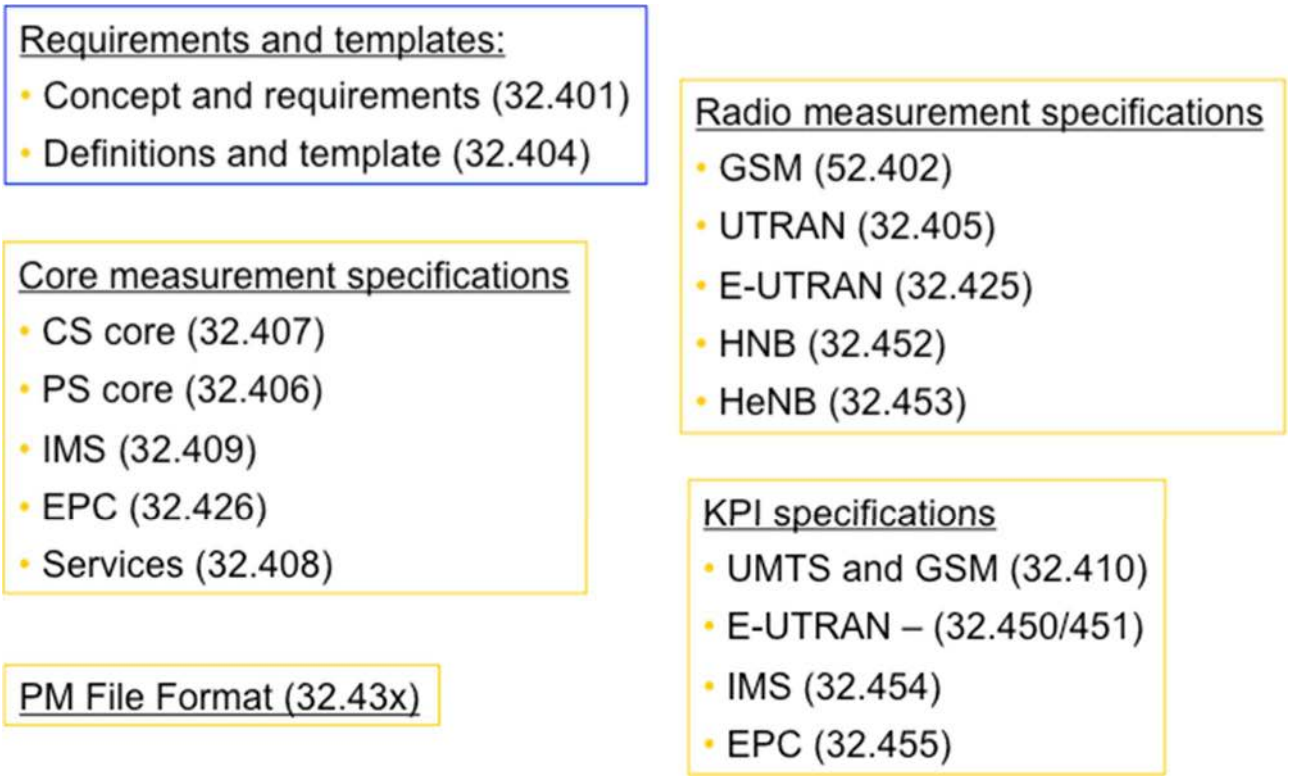


Figure 7.1.1-1: PM-related Specifications 3GPP R10

7.1.2 Performance Measurement Specifications

PS core Performance Measurements – 32.406 [51]

The purpose of this specification is to define performance measurements specific to Core Network Packet Switched Domain in a UMTS network or combined UMTS/GSM network. The standardized measurements result in comparability of measurement data produced in a multi-vendor network.

CS core Performance Measurements – 32.407[52]

The purpose of this specification is to define performance measurements specific to Core Network Circuit Switched Domain in a UMTS network or combined UMTS/GSM network. The standardized measurements result in comparability of measurement data produced in a multi-vendor network.

Services Performance Measurements – 32.408 [53]The purpose of this specification is to define performance measurements specific to teleservices(categorized by Multimedia Messageing Service, Short Message Service, Multimedia Calls, Voice Call and Intelligent Services) in a UMTS network or combined UMTS/GSM network, . The standardized measurements result in comparability of measurement data produced in a multi-vendor network.

IMS Performance Measurements – 32.409 [54]

The purpose of this specification is to define performance measurements specific to an IMS (IP Multimdedia Subsystem) network. The standardized measurements result in comparability of measurement data produced in a multi-vendor network.

EPC Performance Measurements – 32.426 [63]

The purpose of this specification is to define performance measurements specific to an EPC network or combined EPC/UMTS/GSM network. The standardized measurements result in comparability of measurement data produced in a multi-vendor network.

GSM Performance Measurements – 52.402 [118]

The purpose of this specification is to define performance measurements specific to a GSM system. The standardized measurements result in comparability of measurement data produced in a multi-vendor network.**UTRAN Performance Measurements – 32.405 [50]**

The purpose of this specification is to define performance measurements specific to a UTRAN in UMTS network or combined UMTS/GSM network. The standardized measurements result in comparability of measurement data produced in a multi-vendor network.

E-UTRAN Performance Measurements – 32.425 [62]

The purpose of this specification is to define performance measurements specific to an E-UTRAN network. The standardized measurements result in comparability of measurement data produced in a multi-vendor network.**HNB Performance Measurements – 32.452 [71]**

The purpose of this specification is to define performance measurements specific to Home NodeB Subsystem(consists of HNB and HNB-GW). The standardized measurements result in comparability of measurement data produced in a multi-vendor network.

HeNB Performance Measurements – 32.453 [72]

The purpose of this specification is to define performance measurements specific to Home enhanced NodeB Subsystem(consists of HeNB and optionally HeNB-GW). The standardized measurements result in comparability of measurement data produced in a multi-vendor network.

UMTS & GSM KPI's – 32.410 [55]

The purpose of this specification is to define Key Performance Indicators (KPIs) for GSM and UMTS. KPI definitions include high level KPIs that are a) common across GSM and UMTS networks; and b) **specific** to network techniques such as GSM and UMTS networks.

E-UTRAN KPI's – 32.451 [70]

This specification defines requirements (business level requirements, specification level requirements and use case descriptions) related to KPIs for E-UTRAN.

IMS KPI's – 32.454 [73]

This specification defines KPIs for the IP Multimedia Subsystem (IMS).

EPC KPI's – 32.455 [74]

This specification defines KPIs for the Evolved Packet Core(EPC).network.

PM File Format – 32.432 [64], 32.435 [65], 32.436 [66]

This set of specifications describe the general semantics of performance measurement result and collection. It defines the report file format, report file conventions, and the file transfer procedure.

7.2 Trace Specifications

7.2.1 Overview and relationships to IRP Framework

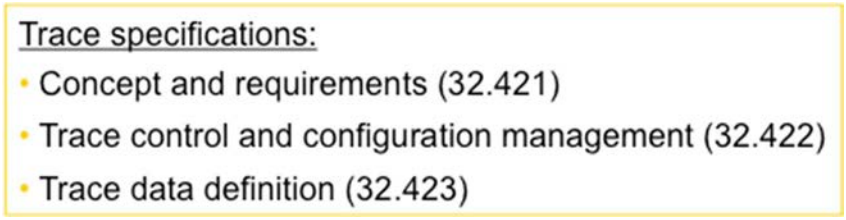


Figure 7.2.1-2: Trace-related Specifications 3GPP R10

7.2.2 Trace Specifications

Trace Control & Configuration – 32.422 [60]

This specification describes the mechanisms used for the control and configuration of the Trace functionality at the EMs, NEs and UEs. It covers the triggering events for starting/stopping of subscriber/UE activity traced over 3GPP standardized signalling interfaces, the types of trace mechanisms, configuration of a trace, level of detail available in the trace data, the generation of Trace results in the Network Elements (NEs) and User Equipment (UE) and the transfer of these results to one or more EM(s) and/or Network Manager(s) (NM(s)).

Trace Data Definitions – 32.423 [61]

This specification describes Trace data definition and management. It covers the trace records content, their format and transfer.

8 Relationships between IRPs

Relationships between Interface IRPs:

- All Interface IRPs utilizing the functionality of the Generic IRP
- All Notification emitting Interface IRPs utilizing the functionality of the Notification IRP

Relationships between NRM IRPs:

- For FMC-enabled NRM IRP, the Generic NRM IRP is utilizing functionality of the FMC FNIM Umbrella Information Model (UIM)
- All NRM IRPs utilizing the functionality of the Generic NRM IRP
- All function-specific NRM IRPs (e.g. Core NRM IRP) utilizing the functionality of the Inventory NRM IRP

9 SON Functions

In order to reduce the operating expenses (OPEX) associated with the management of increasing number of nodes from more than one vendor the concept of the Self-Organizing Network (SON) was introduced. Automation of some network planning, configuration and optimisation processes via the use of SON functions can help the network operator to reduce OPEX by reducing manual involvement in such tasks.

SON functions are Radio Access Technology (RAT) dependent. Generally, SON functions are applicable to E-UTRAN network, and can be applied to UTRAN network on a case by case basis. Currently the following specifications have identified the use cases and requirements in context of SON for standardization:

- Self-establishment of an eNodeB: Related requirements are defined in TS 32.501 [76];
- Automatic Neighbour Relation Management: Related requirements are defined in TS 32.511 [79];
- Self-Optimisation Management: Related requirements are defined in TS 32.521 [80] or TS 28.627 [125];
- Self-Healing Management: Related requirements are defined in TS 32.541 [86];
- Energy Saving Management: Related requirements are defined in TS 32.551 [87];
- Minimization of Drive Tests: Related requirements are defined in TS 32.421 [59] and TS 32.441 [67].

Each SON use case can involve multiple IRPs when corresponding management scenarios are applied. As to IRP solution fulfilment, Self-Optimization Management, Self-Healing Management and Energy Saving Management share the common information model of SON policy NRM (TS 32.522 [81]), while each one also has RAT-dependent information model in E-UTRAN NRM (TS 32.761 [115], 32.762 [116], 32.766 [117],28.657 [137], 28.658 [138], 28.659 [139]) or UTRAN NRM (TS 32.641 [101], 32.642 [102], 32.646 [103] /28.651 [131], 28.652 [132], 28.653 [133]). The IRP relationships of SON functions and measurement definitions are addressed case by case as follows,

Self-establishment of an eNodeB

Interface IRPs:

- SC IRP (32.500 [75], 32.501 [76], 32.502 [77], 32.506 [78])
- Software Management IRP (32.531 [83], 32.532 [84], 32.536 [85])
- Generic IRP (32.311 [25], 32.312 [26], 32.316 [27])
- Alarm IRP (32.111-1 [5], 32.111-2 [6], 32.111-6 [7])
- Test Management IRP (32.321 [28], 32.322 [29], 32.326 [30])
- Basic CM IRP (32.600 [88], 32.601 [89], 32.602 [90], 32.606 [91]) or Bulk CM IRP (32.611 [92], 32.612 [93], 32.616 [94])
- Kernel CM IRP (32.661 [107], 32.662 [108])
- Notification IRP (32.300 [21], 32.301 [22], 32.302 [23], 32.306 [24])
- File Transfer IRP (for ARCF data download) (32.341 [34], 32.342 [35], 32.346 [36])

NRM IRPs:

- Umbrella Information Model (28.620 [121])
- Generic NRM IRP (32.621 [95], 32.622 [96], 32.626 [97] or 28.621 [122], 28.622 [123], 28.623 [124])
- E-UTRAN NRM IRP (32.761 [115], 32.762 [116], 32.766 [117] or 28.657 [137], 28.658 [138], 28.659 [139])

Automatic Neighbour Relation Management

Interface IRP:

- Basic CM IRP (32.600 [88], 32.601 [89], 32.602 [90], 32.606 [91]) or Bulk CM IRP (32.611 [92], 32.612 [93], 32.616 [94])
- Alarm IRP (32.111-1 [5], 32.111-2 [6], 32.111-6 [7])
- Kernel CM IRP (32.661 [107], 32.662 [108])
- Notification IRP (32.300 [21], 32.301 [22], 32.302 [23], 32.306 [24])

NRM IRP:

- Generic NRM IRP (32.621 [95], 32.622 [96], 32.626 [97] or 28.620 [121], 28.621 [122], 28.622 [123], 28.623 [124])
- E-UTRAN NRM IRP (32.761 [115], 32.762 [116], 32.766 [117] or 28.657 [137], 28.656 [136], 28.659 [139])
- UTRAN NRM IRP (32.641 [101], 32.642 [102], 32.646 [103])

Self-Optimisation Management

Interface IRP:

- Basic CM IRP (32.600 [88], 32.601 [89], 32.602 [90], 32.606 [91]) or Bulk CM IRP (32.611 [92], 32.612 [93], 32.616 [94])
- Alarm IRP (32.111-1 [5], 32.111-2 [6], 32.111-6 [7])
- PM IRP (32.410 [55] , 32.411 [56] , 32.412 [57] , 32.416 [58])
- File Transfer IRP (for PM data upload) (32.341 [34], 32.342 [35], 32.346 [36])
- Kernel CM IRP (32.661 [107], 32.662 [108])
- Notification IRP (32.300 [21], 32.301 [22], 32.302 [23], 32.306 [24])

NRM IRP:

- SON Policy NRM IRP (32.521 [80], 32.522 [81], 32.526 [82] or 28.627 [125], 28.628 [126], 28.629 [127])
- Generic NRM IRP (32.621 [95], 32.622 [96], 32.626 [97] or 28.620 [121], 28.621 [122], 28.622 [123], 28.623 [124])
- E-UTRAN NRM IRP (32.761 [115], 32.762 [116], 32.766 [117] or 28.657 [137], 28.656 [136], 28.659 [139])

Measurement Definitions:

- E-UTRAN Performance Measurements (32.425 [62])

Self-Healing Management

Interface IRP:

- Basic CM IRP (32.600 [88], 32.601 [89], 32.602 [90], 32.606 [91]) or Bulk CM IRP (32.611 [92], 32.612 [93], 32.616 [94])
- Alarm IRP (32.111-1 [5], 32.111-2 [6], 32.111-6 [7])
- Kernel CM IRP (32.661 [107], 32.662 [108])
- Notification IRP (32.300 [21], 32.301 [22], 32.302 [23], 32.306 [24])

NRM IRP:

- SON Policy NRM IRP (32.521 [80], 32.522 [81], 32.526 [82] or 28.627 [125], 28.628 [126], 28.629 [127])

- Generic NRM IRP (32.621 [95], 32.622 [96], 32.626 [97] or 28.620 [121], 28.621 [122], 28.622 [123], 28.623 [124])
- E-UTRAN NRM IRP (32.761 [115], 32.762 [116], 32.766 [117] or 28.657 [137], 28.656 [136], 28.659 [139])

Energy Saving Management

Interface IRP:

- Basic CM IRP (32.600 [88], 32.601 [89], 32.602 [90], 32.606 [91]) or Bulk CM IRP (32.611 [92], 32.612 [93], 32.616 [94])
- PM IRP (32.410 [55] , 32.411 [56] , 32.412 [57] , 32.416 [58])
- File Transfer IRP (for PM data upload) (32.341 [34], 32.342 [35], 32.346 [36])
- Kernel CM IRP (32.661 [107], 32.662 [108])
- Notification IRP (32.300 [21], 32.301 [22], 32.302 [23], 32.306 [24])

NRM IRP:

- SON Policy NRM IRP (32.521 [80], 32.522 [81], 32.526 [82] or 28.627 [125], 28.628 [126], 28.629 [127])
- Generic NRM IRP (32.621 [95], 32.622 [96], 32.626 [97] or 28.620 [121], 28.621 [122], 28.622 [123], 28.623 [124])
- E-UTRAN NRM IRP (32.761 [115], 32.762 [116], 32.766 [117] or 28.657 [137], 28.656 [136], 28.659 [139])
- UTRAN NRM IRP (32.641 [101], 32.642 [102], 32.646 [103])

Measurement Definitions:

- E-UTRAN Performance Measurements (32.425 [62])
- UTRAN Performance Measurements (32.405 [50])

Minimization of Drive Tests

Interface IRP:

- Trace Management IRP (32.441 [67], 32.442 [68], 32.446 [69])
- Trace Control and Configuration Management IRP (32.421 [59]. 32.422 [60], 32.423 [61])
- Basic CM IRP (32.600 [88], 32.601 [89], 32.602 [90], 32.606 [91]) or Bulk CM IRP (32.611 [92], 32.612 [93], 32.616 [94])
- Kernel CM IRP (32.661 [107], 32.662 [108])

NRM IRP:

- Generic NRM IRP (32.621 [95], 32.622 [96], 32.626 [97] or 28.620 [121], 28.621 [122], 28.622 [123], 28.623 [124])
- UTRAN NRM IRP (32.641 [101], 32.642 [102], 32.646 [103] or 28.651 [131], 28.652 [132], 28.653 [133])
- E-UTRAN NRM IRP (32.761 [115], 32.762 [116], 32.766 [117] or 28.657 [137], 28.656 [136], 28.659 [139])

Annex A (Informative): IRP Packaging Suggestions

A.1 Overview & Purpose

This Annex provides IRP packaging suggestions, intended to guide service providers as well as equipment and solutions providers to identify and choose suitable standardized management interface capabilities.

Note: No IRP Packaging Suggestions are identified for the present document.

Annex B (normative): Features supported

B.1 Converged management support table

Table B.1.1 below illustrates the "Converged Management" feature support by the individual TS. The intersection of table rows (TS numbers) and columns (features) contains the 3GPP Release number starting from where this feature is supported.

Table B.1.1: Converged management feature support

TS number	Converged management support
32.101	Rel-11
32.103	Rel-11
32.150	Rel-11
32.107	Rel-11
32.156	Rel-11
32.157	Rel-11
32.300	Rel-11
28.390	Rel-12
28.611	Rel-12
28.612	Rel-12
28.616	Rel-12
28.620	Rel-11
28.621	Rel-11
28.622	Rel-11
28.623	Rel-11
28.624	Rel-11
28.625	Rel-11
28.626	Rel-11
28.627	Rel-11
28.628	Rel-11
28.629	Rel-11
28.631	Rel-11
28.632	Rel-11
28.633	Rel-11
28.651	Rel-11
28.652	Rel-11
28.653	Rel-11
28.654	Rel-11
28.655	Rel-11
28.656	Rel-11
28.657	Rel-11
28.658	Rel-11
28.659	Rel-11
28.661	Rel-11
28.662	Rel-11
28.663	Rel-11
28.671	Rel-11
28.672	Rel-11
28.673	Rel-11
28.674	Rel-11
28.675	Rel-11
28.676	Rel-11
28.701	Rel-11
28.702	Rel-11
28.703	Rel-11
28.704	Rel-11
28.705	Rel-11
28.706	Rel-11
28.707	Rel-11
28.708	Rel-11
28.709	Rel-11
28.731	Rel-11
28.732	Rel-11
28.733	Rel-11
28.734	Rel-11
28.735	Rel-11
28.736	Rel-11
28.751	Rel-11
28.752	Rel-11
28.753	Rel-11

B.2 Network sharing management support table

There are several use cases and scenarios for network sharing. Table B.2-1 identifies the list of specifications and the status of support with regards to network sharing that satisfy the 3GPP Stage 1 requirements as defined in the present document.

Legend used in the table:

- **Not Applicable (N/A):**
  The specification is not relevant to network sharing: example TS 32.581, TS 32.582.

- **Not supported: Not supported and no proposal to modify in this release:**
  Needs modifications but might/will not be changed in this release.

- **Not required in this release:** The network sharing scenarios and requirements in the present document do not require support in **this release**.

- **Supported in this release:**
  Specification modified to support network sharing scenarios.

- **Supported with no modification:**
  The specification supports network sharing scenarios with no modifications. Examples are generic NRM.
  Even though this NRM is needed for supporting configuration management for a shared node, there is no change needed to the specification.

- **TBD**: Evaluation in progress.

NOTE: Table B.2-1 is a skeleton and will be populated as the network sharing work is progressing.

The intention of table B.2-1 is to cover all the SA5 Technical Specifications (TS) with an evaluation remark on support for network sharing. If a TS does not appear in the table, it implies that the TS has not been evaluated for impact to support network sharing.

Table B.2-1: Network sharing feature support

3GPP TS Number	3GPP specification Title	Support of network sharing scenarios
28.403	Performance Management; performance measurements for WLAN	Not required in this release. Current SA2 standard does not support WLAN from different operators to be connected to the same CN nodes.
28.601/602/606	Core Network (CN) and non-3GPP access Interworking System Network Resource Model (NRM); Integration Reference Point (IRP);	Not required in this release. Current SA2 standard does not support WLAN from different operators to be connected to the same CN nodes
28.611/612/616	Evolved Packet Core (EPC) and non-3GPP access Interworking System Network Resource Model (NRM); Integration Reference Point (IRP);	Not required in this release. Current SA2 standard does not support WLAN from different operators be connected to the same CN nodes
28.620	Telecommunication management; Fixed Mobile Convergence (FMC) Federated Network Information Model (FNIM) Umbrella Information Model (UIM)	Supported with no modification
28.621/622/623	Telecommunication management; Generic Network Resource Model (NRM) Integration Reference Point (IRP); Requirements	Supported with no modification
28.624/625/626	State management data definition Integration Reference Point (IRP)	Supported with no modification
28.627/628/629	Self-Organizing Networks (SON) Policy Network Resource Model (NRM) Integration Reference Point (IRP);	Not required in this release.
28.631/632/633	Inventory Management (IM) Network Resource Model (NRM) Integration Reference Point (IRP)	Not Applicable
28.651/652/653	Universal Terrestrial Radio Access Network (UTRAN) Network Resource Model (NRM) Integration Reference Point (IRP)	Supported in this release.
28.654/655/656	GSM/EDGE Radio Access Network (GERAN) Network Resource Model (NRM) Integration Reference Point (IRP);	Supported in this release.
28.657/658/659	Evolved Universal Terrestrial Radio Access Network (E-UTRAN) Network Resource Model (NRM) Integration Reference Point (IRP)	Supported in this release.
28.661/662/663	Generic Radio Access Network (RAN) Network Resource Model (NRM) Integration Reference Point (IRP);	Supported with no modification
28.671/672/673	Home Node B (HNB) Subsystem (HNS) Network Resource Model (NRM) Integration Reference Point (IRP)	Not Applicable (no plan to apply network sharing concepts to Home Node B (HNB)).

28.674/675/676	Home enhanced Node B (HeNB) Subsystem (HeNS) Network Resource Model (NRM) Integration Reference Point (IRP)	Not Applicable (no plan to apply network sharing concepts to Home enhanced Node B (HeNB)).
28.701/702/703	Telecommunication management; Core Network (CN) Network Resource Model (NRM) Integration Reference Point (IRP)	See 32.631/2/6. Not required in this release (Impact of GWCN, e.g. sharing of SGSN, on existing specs not addressed in this release).
28.704/705/706	Telecommunication management; IP Multimedia Subsystem (IMS) Network Resource Model (NRM) Integration Reference Point (IRP); Requirements	Not required in this release.
28.707/708/709	Evolved Packet Core (EPC) Network Resource Model (NRM) Integration Reference Point (IRP)	Not required in this release (Impact of GWCN, i.e. sharing of MME, on existing specs not addressed in this release).
28.731/732/733	Transport Network (TN) interface Network Resource Model (NRM) Integration Reference Point (IRP)	Not Applicable (no plan to apply network sharing concepts to transport network).
28.734/735/736	Signalling Transport Network (STN) interface Network Resource Model (NRM) Integration Reference Point (IRP);	Not Applicable (no plan to apply network sharing concepts to signalling transport network).
28.751/752/753	Subscription Management (SuM) Network Resource Model (NRM) Integration Reference Point (IRP); Requirements	Not Applicable
32.101	Principles and high level requirements	Supported with no modification.
32.102	Architecture	Supported with no modification.
32.103	Integration Reference Point (IRP) overview and usage guide	Supported with no modification
32.107	Telecommunication management; Fixed Mobile Convergence (FMC) Federated Network Information Model (FNIM)	Supported with no modification
32.111-1/2/3/6	Fault Management;: 3G fault management requirements	TBD
32.121/122/126	Advanced Alarm Management (AAM) Integration Reference Point (IRP):	TBD
32.140	Telecommunication management; Subscription Management (SuM) requirements	Not Applicable
32.141	Telecommunication management; Subscription Management (SuM) architecture	Not Applicable
32.150	Integration Reference Point (IRP) Concept and definitions	Supported with no modification
32.151	Integration Reference Point (IRP) Information Service (IS) template	Supported with no modification
32.152	Integration Reference Point (IRP) Information Service (IS) Unified Modelling Language (UML) repertoire	Supported with no modification

32.153	Integration Reference Point (IRP) technology specific templates, rules and guidelines	Supported with no modification
32.154	Backward and Forward Compatibility (BFC); Concept and definitions	Not Applicable
32.155	Requirements template	Supported with no modification
32.156	Fixed Mobile Convergence (FMC) model repertoire	Supported with no modification
32.157	Telecommunication management; Integration Reference Point (IRP) Information Service (IS) template	Supported with no modification
32.181/182	Telecommunication management; User Data Convergence (UDC); Framework for Model Handling and Management	Not Applicable (no plan to apply network sharing concepts to UDC).
32.300	Configuration Management (CM); Name convention for Managed Objects	TBD
32.301/302/306	Configuration Management (CM); Notification Integration Reference Point (IRP)	Supported with no modification
32.311/312/316	Generic Integration Reference Point (IRP) management	Supported with no modification
32.321/322/326	Test management Integration Reference Point (IRP);	Supported with no modification
32.331/332/336	Notification Log (NL) Integration Reference Point (IRP)	Supported with no modification
32.351/352/356	Communication Surveillance (CS) Integration Reference Point (IRP)	Not Applicable
32.361/362/366	Entry Point (EP) Integration Reference Point (IRP); Requirements	Supported with no modification
32.371/372/376	Telecommunication management; Security Management concept and requirements	Supported with no modification
32.381/382/386	Telecommunication management; Partial Suspension of Itf-N Integration Reference Point (IRP); Requirements	Supported with no modification
32.391/392/396	Telecommunication management; Delta synchronization Integration Reference Point (IRP); Requirements	Supported with no modification
32.401	Telecommunication management; Performance Management (PM); Concept and requirements	Supported with no modification
32.404	Telecommunication management; Performance Management (PM); Performance measurements; Definitions and template	Supported with no modification

32.405	Telecommunication management; Performance Management (PM); Performance measurements; Universal Terrestrial Radio Access Network (UTRAN)	TBD
32.406	Telecommunication management; Performance Management (PM); Performance measurements; Core Network (CN) Packet Switched (PS) domain	Supported with no modification
32.407	Telecommunication management; Performance Management (PM); Performance measurements; Core Network (CN) Circuit Switched (CS) domain; UMTS and combined UMTS/GSM	Supported with no modification
32.408	Telecommunication management; Performance Management (PM); Performance measurements; Teleservice	Not Applicable
32.409	Telecommunication management; Performance Management (PM); Performance measurements; IP Multimedia Subsystem (IMS)	Not required in this release.
32.410	Telecommunication management; Key Performance Indicators (KPI) for UMTS and GSM	TBD
32.411/412/416	Telecommunication management; Performance Management (PM) Integration Reference Point (IRP): Requirements	Supported with no modification
32.421/422/423	Telecommunication management; Subscriber and equipment trace;	TBD
32.425	Telecommunication management; Performance Management (PM); Performance measurements Evolved Universal Terrestrial Radio Access Network (E-UTRAN)	TBD
32.426	Telecommunication management; Performance Management (PM); Performance measurements Evolved Packet Core (EPC) network	TBD
32.432	Telecommunication management; Performance measurement: File format definition	TBD
32.435	Telecommunication management; Performance measurement; eXtensible Markup Language (XML) file format definition	TBD
32.436	Telecommunication management; Performance measurement: Abstract Syntax Notation 1 (ASN.1) file format definition	TBD

32.441/442/446	Trace Management Integration Reference Point (IRP)	TBD
32.450	Key Performance Indicators (KPI) for Evolved Universal Terrestrial Radio Access Network (E-UTRAN): Definitions	TBD
32.451	Key Performance Indicators (KPI) for Evolved Universal Terrestrial Radio Access Network (E-UTRAN); Requirements	TBD
32.452	Performance Management (PM); Performance measurements Home Node B (HNB) Subsystem (HNS)	Not Applicable
32.453	Performance Management (PM); Performance measurements Home enhanced Node B (HeNB) Subsystem (HeNS)	Not Applicable
32.454	Key Performance Indicators (KPI) for the IP Multimedia Subsystem (IMS); Definitions	Not required in this release.
32.455	Key Performance Indicators (KPI) for the Evolved Packet Core (EPC); Definitions	Not required in this release.
32.500	Self-Organizing Networks (SON); Concepts and requirements	Not required in this release.
32.501/502/506	Self-configuration of network elements; Concepts and requirements	Supported with no modification
32.511	Automatic Neighbour Relation (ANR) management;	TBD (handling of ANR may be impacted by RAN sharing)
521/522/526	Self-Organizing Networks (SON) Policy Network Resource Model (NRM) Integration Reference Point (IRP);	Not required in this release.
32.531/532/536	Software management (SwM); Concepts and Integration Reference Point (IRP)	Not Applicable
32.541	Self-Organizing Networks (SON); Self-healing concepts and requirements	Not Applicable
32.551	Energy Saving Management (ESM); Concepts and requirements	Not Applicable
32.571/572	Home Node B (HNB) and Home eNode B (HeNB) management; Type 2 interface	Not Applicable
32.581/582/583/584	Home Node B (HNB) Operations, Administration, Maintenance and Provisioning (OAM&P): Type 1 interface	Not Applicable
32.591/582/593/594	Home enhanced Node B (HeNB) Operations, Administration, Maintenance and Provisioning (OAM&P): Type 1 Interface	Not Applicable
32.600/601/602/606	Configuration Management (CM)	Supported with no modification
32.611/612/616	Configuration Management (CM); Bulk CM Integration Reference Point (IRP)	Supported with no modification

32.621/622/626	Configuration Management (CM); Generic network resources Integration Reference Point (IRP)	Not required in this release.
32.631/632/636	Configuration Management (CM); Core network resources Integration Reference Point (IRP)	Supported with no modification
32.661/662/666	Configuration Management (CM); Kernel CM	Supported with no modification
32.671/672/676	Configuration Management (CM); State Management Integration Reference Point (IRP)	Not Applicable
32.690	Telecommunication management; Inventory Management (IM); Requirements	Not Applicable
32.691/692/696	Inventory Management (IM) Network Resource Model (NRM) Integration Reference Point (IRP)	Not Applicable

Annex C (informative): Change history

Change history							
Date	**TSG #**	**TSG Doc.**	**CR**	**Rev**	**Subject/Comment**	**Old**	**New**
2010-12	SP-50	SP-100764	--	--	Submitted to SA#50 for Information	0.1.0	1.0.0
2011-05	SP-52	SP-110272	--	--	Submitted to SA#52 for Approval	1.2.0	2.0.0
2011-06	SP-52	--	--	--	Publication	2.0.0	10.0.0
2012-06	SP-56	SP-120370	001	2	Add relation description between IRPs	10.0.0	11.0.0
2012-12	SP-58	SP-120783	003	2	CR R11 32.103 Add usage description for SON functions	11.0.0	11.1.0
2013-03	SP-59	SP-130060	004	2	Enhancements for Converged Management	11.1.0	11.2.0
2013-09	SP-61	SP-130433	005	1	correction of references	11.2.0	11.3.0
2014-06	SP-64	SP-140332	006	2	Add missing feature support matrix	11.3.0	11.4.0
		SP-140359	007	-	remove the feature support statements		
2014-10					Automatic upgrade (MCC)	11.4.0	12.0.0
2014-12	SP-66	SP-140798	009	-	Remove the mapping information between 32 series and 28 series	12.0.0	12.1.0
			010	-	Add feature support statements for Rel-12		
		SP-140797	012	-	Update feature support statement for TS 32.300		
		SP-140800	013	2	Add annex for network sharing management support table		

Change history							
Date	**Meeting**	**TDoc**	**CR**	**Rev**	**Cat**	**Subject/Comment**	**New version**
2020-09	SA#89e	SP-200728	0021	-	A	Correction of reference list and adding references throughout the document	12.2.0
2020-12	SA#90e	SP-201048	0027	1	A	Correction of reference list and adding references in the document	12.3.0

History

Document history		
V12.0.0	October 2014	Publication
V12.1.0	January 2015	Publication
V12.2.0	November 2020	Publication
V12.3.0	January 2021	Publication